Logic, Language, and Philosophy

Proposition logic: syntax

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Today

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Introduction

Connectives

Proposition logic is the logic of connectives

Connectives are expressions that combine sentences into new sentences:

- *'and', 'or', 'but', 'provided that', 'but only if', ...*
- *'it is not the case that', 'it might be the case that', 'it used to be the case that', ...*

Using connectives

Start with basic sentences, i.e., sentences that do not contain connectives:

- John stays at home
- John goes to the office
- John will be home late
- Beijing is a big city
- Shanghai is a big city
- ...

Build complex ones by combining them using connectives:

- John stays at home or he goes to the office
- Beijing is a big city and so is Shanghai
- If John goes to the office, then he will be home late
- Beijing is not a big city
- ...

Truth-functional connectives

Compare:

1. John stays home and Mary goes to the office
2. John stays home because Mary goes to the office

Observation:

- If 2 is true then so is 1, but not the other way around

Explanation:

- The truth of 1 only depends on the truth of its component parts, but the truth of 2 requires more
- In other words: 'and' is a truth-functional connective, 'but' is not

A connective $\circ$ is truth-functional if the truth value of $A \circ B$ is completely determined by the truth values of A and B

Non truth-functional connectives

Examples:

- ‘but’, ‘because’, ‘despite’, ...
- ‘necessarily, ‘it is possible that, ‘it is obligatory that’, ‘it will be the case that’, ...

Note:

- Not truth-functional $\neq$ no logic:
  modal logic, deontic logic, tense logic, ...
- *Some* not truth-functional aspects need to be explained in other ways (pragmatics)

Logic of truth-functional connectives

The logic of truth-functional connectives can be discerned in the validity of *patterns* such as:

- From the truth of '*A* and *B*', we can conclude that '*A*' is true and that '*B*' is true
- Conversely: if '*A*' is true and '*B*' is as well, then we can conclude that '*A* and *B*' is true
- From the truth of '*A* or *B*', we cannot conclude that '*A* 'is true , nor can we conclude that ''*B* is true; but we can conclude that it is not the case that neither '*A*' nor '*B*' is true
- From the truth of 'If *A* then *B*' and the truth of 'It is not the case that *A*', we can conclude that '*B*' is false
- ...

Sentences and propositions

In order to describe the logic of connectives, we need to abstract away from all the actual content of concrete sentences, except for the connectives that they contain

So:

- The contents of sentences are called *propositions*
- Propositions are the variables of a system of propositional logic
- The connectives that combine propositions are its logical constants

Propositional languages

Vocabulary

Definition (Vocabulary)

VOC_L, the vocabulary of L, consists of:

1. An infinite set of proposition letters $p_1, p_2, \ldots p_n \ldots$
2. A set of connectives: $\{\neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$
3. Brackets: (,)

Note:

- Instead of $p_1, p_2, \ldots p_n \ldots$, we also write $p, q, r, s \ldots$.
- Important: proposition letters can be enumerated
- Proposition letters are also called 'atomic formulas', or 'atoms', for short
- We use connectives to form complex formulas

Connectives of proposition logic

Connective	*Composite formula*	*Meaning*
$\neg$: negation	$\neg p$: negation of p	it is not the case that p
$\wedge$: conjunction	$p \wedge q$: conjunction of p and q	p and q
$\vee$: disjunction	$p \vee q$: disjunction of p and q	p or q
$\rightarrow$: implication	$p \rightarrow q$: (material) implication of p and q	if p, then q
$\leftrightarrow$: equivalence	$p \leftrightarrow q$: (material) equivalence of p and q	p if and only if q

Table 1: Connectives and their meanings

Terminology:

- conjunction: conjuncts
- disjunctions: disjuncts
- implication: antecedent, consequent

Syntax

Definition (Syntax)

The formulas of L, given VOC_L, is the set of expressions:

1. All proposition letters in VOC_L are formulas of L
2. If ϕ is a formula of L, then so is $\neg\phi$
3. If ϕ and ψ are formulas of L, then so are $(\phi \wedge \psi)$, $(\phi \vee \psi)$, $(\phi \to \psi)$, and $(\phi \leftrightarrow \psi)$
4. Only that which can be generated by the clauses (1)-(3) in a finite number of steps is a formula of L

Note:

- We leave out the outermost pair of brackets and write, e.g., $p \to (q \vee r)$ instead of $(p \to (q \vee r))$
- 'Meta-variables' $\phi, \psi, \chi, \ldots$ are not formulas, but are used to *talk about* formulas:

Examples

Formulas:

- p
- $((p \to q) \land p) \to q$
- $\neg p \land ((s \leftrightarrow t) \leftrightarrow ((q \lor p) \to \neg t))$

Not formulas:

- q
- $p \lor q \to \neg r$
- $((s \to t))$

Construction trees

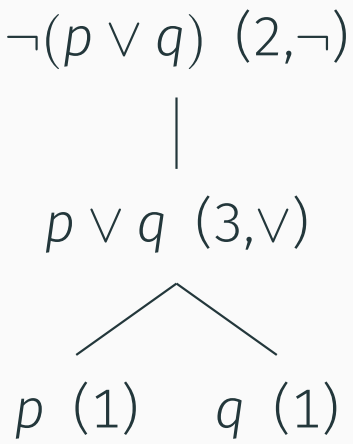

Figure 1: Construction tree $\neg(p \vee q)$

Construction trees (2)

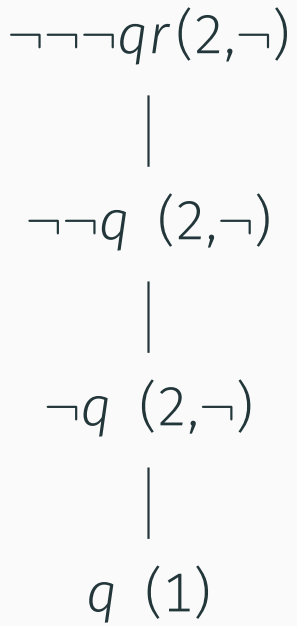

Figure 2: Construction tree $\neg\neg\neg q$

Construction trees (3)

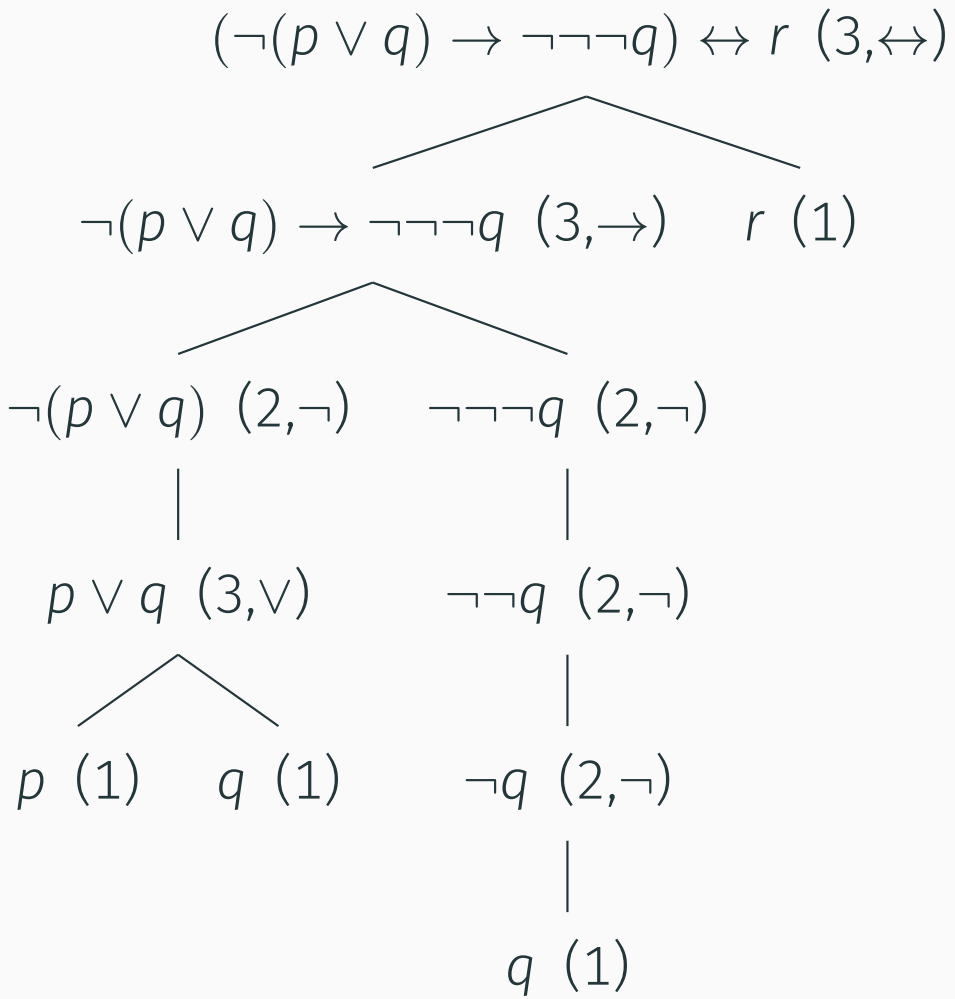

Figure 3: Construction tree $(\neg(p \vee q) \to \neg\neg\neg q) \leftrightarrow r$

Induction

Proof by induction

An inductive proof has the following structure:

- Base case
- Induction step(s) (a.k.a. 'induction hypothesis')

It uses an inductive definition of the domain that it proves something about:

Example: the natural numbers $\mathbb{N}$ are defined inductively as:

- $0 \in \mathbb{N}$
- if $n \in \mathbb{N}$, then $n + 1 \in \mathbb{N}$

Proof of 'Every number $n \geq 0$':

- Base step: $0 \geq 0$, since $0 = 0$
- Induction step: suppose $k \geq 0$
- Then $k + 1 \geq 0$ since :
  - $k + 1 \geq k$ and
  - for all m, n, k: if $m \geq n$ and $n \geq k$, then $m \geq k$

Proof by induction on complexity

The syntax of proposition language is also given by an inductive definition:

- Base step: proposition letters are formulas
- Inductions steps (plural!): the clauses that state how the connectives form new formulas

Proof by induction on construction has the same form:

- Base step: show that something holds for the propositions letters
- Induction steps: shows that if it holds for a given formula or two given formulas, it holds for the formulas that are constructed by means of each of the connectives

Example

All formulas ϕ are built from a finite set of proposition letters

Proof:

- Base step: a proposition letter consists of itself, which forms a finite set
- Induction steps:
  - Suppose ϕ is built from a finite set of proposition letters, then so does $\neg\phi$, since negation does not introduce new proposition letters
  - Suppose ϕ and ψ both are built from finite sets of proposition letters, then so does $\phi \wedge \psi$, since conjunction does not introduce new proposition letters and the union of two finite sets is finite
  - [and similarly for the other connectives]

Using a propositional language

Translation and translation key

Procedure:

1. Choose proposition letters for different atomic sentences
2. Identify the logical constants
3. Represent the complete sentence making as much of the logical structure explicit as possible

Step 1 is laid down in a translation key

Examples

Example:

- Sentence:
  If I have lost if I cannot make a move, then I have lost
- Translation key:
  p: I can make a move; q: I have lost
- Translation:
  $(\neg p \to q) \to q$

Examples (2)

Example:

- Argument:
  If the butler didn't do it, it was either the cook or her lover. If the lover did it, he would have run away. But he didn't. So, it's the cook or the butler. Both would have accused the other if they had done it. But the cook did not say anything and the butler did. So it was the butler
- Translation key:
  p: the butler did it; q: the cook did it; r: the cook's lover did it; s: the cook's lover runs away; t: the butler accuses the cook; u: the cook accuses the butler
- Translation:
  $\neg p \to (q \vee r), r \to s, \neg s$; so: $\neg r$ and thus $p \vee q$; $(p \to t) \wedge (q \to u), \neg u$; so $\neg q$; and thus p

Homework

Homework

Make the following exercises from chapter 3:

- Exercise 1: f–j
- Exercise 2: a–b
- Exercise 3
- Exercise 5: d–l
- *Bonus*: exercise 4

Read:

- Chapter 4
- Papers by Prior and Belnap